Primary Distribution Systems

The most frequently used voltages and wiring in the primary distribution are 11 kV, or 22 kV in Egypt.

There are several configurations of the primary feeders:

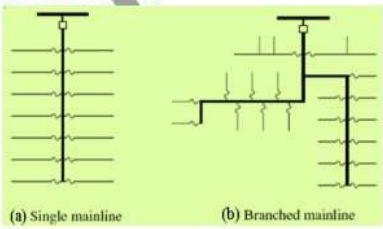
1- A Radial System: **Primary distribution, in low load density areas, is a radial system. A radial system is economical but low reliability.**

Advantages of a radial designs: **(1) Protection is basically overcurrent. (2) Lower fault currents.**

(3) System design is less expensive. (4) Voltage regulation and power flow control are easier.

The general radial circuit design such as the single mainline, or the branched mainline.

A Single primary line. Either a single mainline, or a branched mainline.

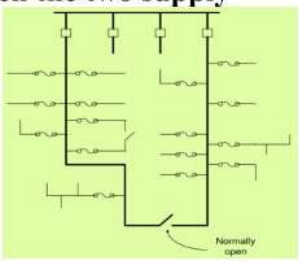


2- The Loop Connection:

When the main feeder is supplied from two independent distribution substations.

In high load density areas use loop connection system.

- **Good reliability (A more economical and fairly reliable arrangement).**
- **The problem with this connection is the circulating current that occurs when the two supply station voltages are different.**
- **The circulating current can be avoided by using the open-loop connection.**
- **The normally open tie-switch connects the feeder to the adjacent distribution substation.**
- ✓ **This configuration presents a greater reliability level than the single feeder configuration.**



3- A Primary Cable Network:

- **In large cities, where the load density is very high, a primary cable network is used.**
- **The high cost of the network limits its application.**

A capacitor bank. **A three-phase switched capacitor bank is rated two-thirds of the total average reactive load and installed two-thirds of the distance out on the feeder from the source.**

- **The capacitor bank improves the power factor and reduces voltage drop at heavy loads.**
- **At light loads, the capacitor is switched off to avoid over-voltages.**
- **Primary distribution circuits are generally radial in design, unlike transmission systems where circuit designs are meshed or looped.**

1. Most primary feeders in rural areas are overhead lines using pole-mounted distribution transformers.
2. In urban areas, an underground cable system is used.
4. The underground system is not affected by weather and is highly reliable.

In Primary distribution systems:

- The three-phase four-wire main feeders supply single-phase lateral feeders.
- A recloser and a sectionalizing switch divide the main feeder into two parts.

Secondary Distribution Systems

The secondary distribution system provides electric energy to the customers through the distribution transformers and secondary cables. High-rise buildings are supplied by a three-phase four-wire spot network.

Radial Configuration, Spot Configuration And Network Configuration.

- The radial configuration is used in rural.
- The spot configuration is used for large loads concentrated in one point (e.g. factories and large buildings).
- The network configuration is used to serve great number of loads distributed over a large area.
- Generally, the secondary network is used in urban areas with high load density.

Feeder Voltage Regulation

- The voltage supplied to each customer on a distribution feeder should be within the ± 5% limit.
  - e.g. for 120 V, the corresponding limits are 114 V and 126 V.
- The voltage drop can be reduced by the application of a shunt capacitor or tape change voltage regulators.
- The capacitor is switched off during the night when the load is light and switched on when the load is heavy.
- Residential underground systems require less shunt capacitance.
- Shunt capacitors are used for power factor correction and loss reduction.
- The concept of optimization includes the minimization of the operation, installation, and investment costs.
- In this case the optimum location of the capacitor bank is around two-thirds (2/3) of the length of the feeder.
- The voltage regulator is a tap-changing transformer, which is located at the supply end of the feeder.

Secondary Circuits: Voltage Drop, Power Loss, and Power Factor

The voltage drop (ΔV) : is defined as the amount of voltage loss that occurs through all or part of a circuit due to its impedance.

Power losses

The power lost in a cable depends on the cable length, cable size and the current through the cable.

Three ways to lower the resistance in the cable:

- **Shorten the length of the cable. ⟹ The cable impedance is reduced**
- **Increase the size of the conductor. ⟹ Resistance is reduced**
- **Decrease the current of load through the cable by reducing the loading level.**

Kinds of Power Losses: ***Definition***

1. Active power losses **which results from the heating of the conductor and the cable (or wire) insulation.**

The active power losses can be simplified to $P_{loss} = I^2R$, where R is the conductor resistance.

2. Reactive power losses **which is the energy needed for creation and storage of the magnetic fields associated with the flow of current, and the electric fields associated with the voltage separation.**

The reactive power loss can be simplified as $Q_{loss} = I^2X$, where X is the reactance of the line. It is well known that $X = 2\pi fL$, where L is the inductance of the conductors.

Reducing the Power Loss: by **Lower Resistance**: **by using Short length cable, with increased size conductor or lower current: by reducing the loading level.**

Definition **The power factor of an AC electrical power system is defined as the ratio of the real power (P) flowing to the load to the apparent power (S) in the circuit, and is a dimensionless number in the closed interval of −1 to 1.**

Power Factor Triangle:

- **Real power: is the instantaneous product of voltage and current and represents the capacity of the electricity for performing work in a particular time.**
- **Apparent power: is the product of the current and voltage of the circuit.**
- **Reactive power: Due to energy stored in the load and returned to the source, or due to a non-linear load that distorts the wave shape of the current drawn from the source.**

$$S = P_L + jQ_L \qquad |S|^2 = P_L^2 + Q_L^2 \qquad |S| = \sqrt{P_L^2 + Q_L^2}$$

$$\text{Power factor} = \cos\phi = \frac{P_L,\ \text{real power}}{S_L,\ \text{apparent power}}$$

Due to the Reactive power, the apparent power will be greater than the real power.

A negative power factor occurs when the device (load) generates power, which then flows back towards the source, which is normally considered the generator.

Lagging, leading and Unity power factors:

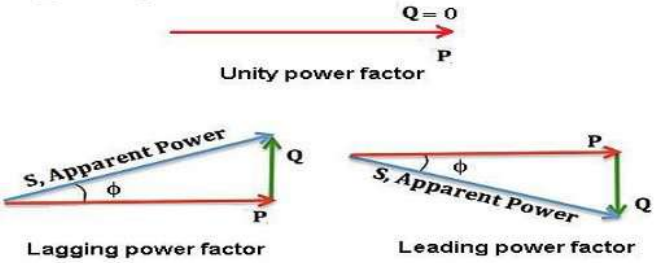
The terms Lagging and leading power factors refer to whether the phase of the current is leading or lagging the phase of the voltage.

- **A lagging power factor** signifies that the load is **inductive**, as the load will **"consume" reactive power**, and therefore **the reactive component Q is positive**.
- **A leading power factor** signifies that the load is **capacitive**, as the load **"supplies" reactive power**, and therefore **the reactive component Q is negative**.
- **A Unity power factors** signifies that the load is **Resistive**, then **reactive power equal "Zero"** and **the apparent power will be equal the real power**.

In industrial and commercial facilities, the majority of electrical equipment acts like resistors or inductors. Resistive loads include incandescent lights, baseboard heaters, and cooking ovens. Inductive loads include fluorescent lights, AC induction motors, arc welders, and transformers.

- **A power factor of less than one means that: the voltage and current waveforms are not in phase. The apparent power will be greater than the real power.**
- **A power factor equal one means that: the voltage and current waveforms are in phase. The apparent power will be equal the real power.**

In an electric power system, a load with a low power factor draws more current than a load with a high power factor for the same amount of useful power transferred. The higher currents increase the energy lost in the distribution system, and require larger wires.



Advantages of improving the power factor:

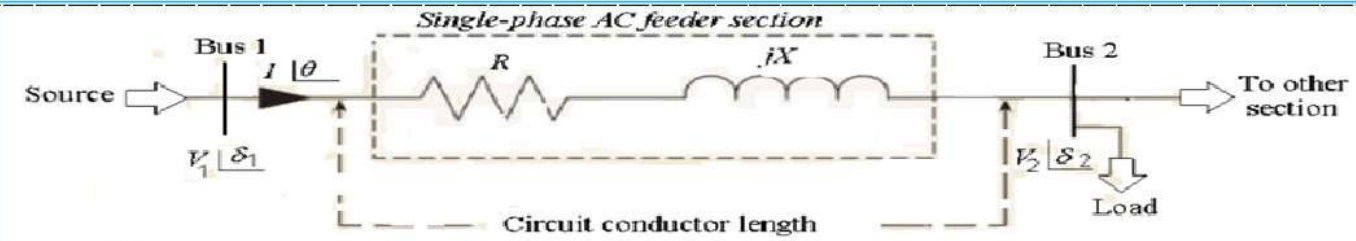
(1) Reduction in circuits current. (2) Increase in voltage level at load. (3) Reduction in investments.

(4) Reduction in system energy losses. (5) Reduction in the kVA loading.

(6) Lower utility operating costs and lower customer costs.

(7) Reduction in kVA demand charges for large consumers.

(8) Improved equipment operation by reducing the voltage drop.



Example: **A wound coil that has an inductance of 180mH and a resistance of 35Ω is connected to a 100V 50Hz supply. Calculate: a) the impedance of the coil, b) the current, c) the power factor, d) the apparent power consumed. e) Draw the resulting power triangle for the above coil.**

Load Characteristics, and Analysis

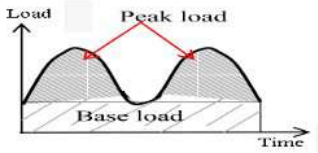
No.	Load Characteristic	Definition
1	Demand	The demand of an installation or system is the load that is drawn from the supply at a specified interval of time; it is expressed in KWs, KVA or Amperes.
2	Connected Load	The connected load is the sum of continuous ratings of all the equipment's connected in supply system. ➢ The load which a power system has to supply is never constant because of variable demands at different time of the day.
3	Load Curve	Load curve: is a graphic record showing the demand of the power for every instant during the hour, the day, the month or the year, which is called the chronological load curve.
4	Maximum load (or the peak load)	Represents the maximum power that consumed by the load during a specific given time. ➢ The peak load equal to the maximum actual power generated by the plant minus the transmission and distribution losses.
5	Average Load	The average load: is the average power that consumed by the load during a certain period of time. $\text{Avarage Load} = \frac{\text{Area under the load curve (KWH)}}{\text{No. of hours (h)}}$ ➢ Average load equal to the average power that generated by the plant during the same period of time.
6	Load Factor	The load factor is the ratio of the average load to the maximum load for a certain period of time. $\text{Load Factor} = \frac{\text{Average Load}}{\text{Max. Load}}$ ➢ The load factor is ≤ 1.
7	Demand Factor	The demand factor is the ratio of the maximum load to the connected load. ➢ The demand factor is ≤ 1.0. ➢ The lower the demand factor, the less system capacity required to serve the connected load.
8	Diversity Factor	The ratio of the sum of individual maximum demands to maximum demand on the power station. ➢ The diversity factor is always ≥ 1.0
9	Utilization Factor(Ku)	The time that a equipment is in use / The total time that it could be in use.

Notes:

- **Ku- factor must be applied to each individual load.**
- **Examples from industry, for motors Ku=0.75, for incandescent-lighting loads, Ku=1.**
- **For socket-outlet circuits, Ku; depend on the type of appliances being supplied from the sockets concerned.**

Base Load and Peak Load on Power Station

- **Base load: The unvarying load which occurs almost the whole day on the station.**
  Peak load: The various peak demand of the load above the base load of the station.



Load Duration curve

- **When the load elements of a load curve are arranged in the order of descending magnitudes.**
- **The area under daily load curve = Area under daily load duration curve = Total energy generated (KWh) on the day.**

Problems

1. If a residence having 6000W equipment connected has a maximum demand of 3300W, then the demand factor = ----------

2. A substation has three outgoing feeders:

Feeder-1 has maximum demand 10 MW at 10:00 am.

Feeder-2 has maximum demand 12 MW at 7:00 pm.

Feeder-3 has maximum demand 15 MW at 9:00 pm. While the maximum demand of all three feeders is 33 MW at 8:00 pm. Then the diversity factor = ----------.

3. A motor may only be used for eight hours a day, 50 weeks a year. The hours of operation would then be 2000 hours, and the motor Utilization factor for a base of 8760 hours per year would be ----------

4. We have four individual feeder-circuits with connected loads of 250 kVA, 200 kVA, 150 kVA and 400 kVA and demand factors of 90%, 80%, 75% and 85% respectively. Use a diversity factor of 1.5 to find the kVA rating of the main distribution feeder and transformer.

5. If the average load 250 Kw and the maximum load is 750 Kw, then find the load factor.

6. Consider a conveyor belt made up of six sections, each driven by a 2 kW motor.the sum of the demand loads is 6 kW but the maximum load presented by the system at any time is only 1.5 kW. Therefore, Diversity factor =----------, Demand Factor =----------

7. An electric motor is used 8 hours a day, 5 days a week, and 50 weeks a year. Then the motor Utilization factor is ----------

8. A distribution feeder serves 5 houses, each house has a peak demand of 5 KW. The feeder peak turns out to be 20 kW. The diversity is then = ----------

Sheet chapter 1

Q1: Choose the correct answer:

1. Underground lines are typically made of: A) Copper conductors. B) Bare aluminum (ACSR) conductors. C) Cable with polymer insulation (XPLE and EPR).	2. The voltage drop can be reduced by the application of a shunt …… A) inductor. B) resistor. C) capacitor.
3. Circuit breakers, reclosers, and fuses are: A) switching devices. B) controlling devices. C) Protection devices.	4. Switching and protection such as-------- A) Relay. B) Recloser. C) Transformer.
5. A potential transformer is a conventional transformer with primary and secondary windings on a …… core. A) separate. B) mutual. C) common.	6. The voltage drop can be reduced by the application of A) Shunt capacitor. B) Bank capacitor. C) Switch capacitor.
7. A circuit breaker is---- A) A power factor correcting device. B) A waveform correcting device. C) A current interrupting device.	8. .. is used to reduce the primary circuit voltage to a safe value (120 V) A) Power transformer. B) Potential transformer. C) Current transformer.
9. .....are installed at intermediate points of long primary lines in order to compensate the voltage drop produced along the circuit. A) Voltage regulators. B) Series reactors. C) All of them.	10. Which of the following considered protection device. A) Relay and fuse. B) Recloser. C) All of them.
11. The distributors for residential areas are A) Single phase. B) Three phase three wire. C) Three phase four wires.	12. Three phase -4 wire system is commonly used on: A) Primary transmission. B) Secondary transmission. C) Secondary distribution.
13. Radial system is …… A) expensive and low reliability. B) expensive and high reliability. C) economical and low reliability.	14. To reduce energy transportation losses in electrical power systems, we use…. A) Step up transformer. B) Step down transformer. C) Both A snd B.
15. A more fairly reliable arrangement is: A) the open loop connection. B) the loop connection. C) the radial system.	16. The value of demand factor--- A) Greater than unity. B) Less than unity. C) Always more than unity.
17. A …… power factor signifies that the load is capacitive. A) leading B) unity C) lagging	18. A …… power factor signifies that the load is resistive. A) leading B) unity C) lagging

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19. The voltage supplied to each customer on a distribution feeder should be within the …..limit. A) ± 3% B) ± 5% C) ± 10%	20. A …… power factor signifies that the load is inductive. A) leading B) unity C) lagging
21. The power factor of an electrical power system is in the closed internal of: A) 0 to +1. B) -1 to +1. C) -1 to 0.	22. A …… power factor signifies that the load is capacitive. A) leading B) unity C) lagging
23. Active power and apparent power are respectively represented by? A) kW and kVAR B) kVA and kVAR C) kW and kVA	24. Lower power factor is usually not due to----- A) Discharge lamp. B) Incandescent lamps C) Induction furnace.
25. Unit of reactive power: A) Volt Ampere. B) Watt. C) VAR.	26. A lagging power factor signifies that the load is: A) capacitive. B) inductive. C) Resistive.
27. A leading power factor signifies that the load is… A) Resistive. B) Inductive. C) Capacitive.	28. In urban areas, an ……..cable system is used. A) Overhead. B) underground. C) open loop.
29. A more economical and fairly reliable arrangement, when the main feeder is supplied from two dependent distribution substations. A) Radial connection. B) loop connection. C) Network connection.	30. When the power factor < 1, this is mean------- A) The real power P less than Apparent power S. B) The real power P greater than Apparent power S. C) The real power P equal Apparent power S.
31. The active power losses can be simplified to------- A) I^2R. B) I^2XL. C) I^2XC.	32. The feeder power losses are of..... losses. A) Active. b) Reactive. C)Active and reactive.
33. In single phase circuits, the active power losses is proportional to.....circuit current. A) Half. B) Double. C) Square.	34. The capacitor bank improves the power factor and reduces voltage drop at ..... loads. A) Light. B) Moderate. C) Heavy.
35. The primary reason for the low power factor is due to the installation of--- A) Induction motors. B) DC motors C) Synchronous motors.	36. ..... represents the maximum power that consumed by the load during a specific given time. A)Peak load. b)Average load. c)Connected load
37. All feeder will cause a voltage drop ……. Vols per unit length. This required voltage regulation. A) IR B) IX c) IZ	38. The voltage drop can be reduced by the application of a …….. capacitor. A) parallel. B) series. C) Not of these.
39. What does the area under the load curve represent? A) System voltage. B) Current. C) Energy consumed.	40. What is the result of the product of diversity factor and maximum demand? A) Average demand. B) Sum of consumers maximum demand. C) Installed capacity.

Q2: Complete these sentences:

1. Circuit breakers, reclosers, and fuses are …….
2. Sectionalizer is a ……
3. A …… is a switching device used to isolate a system element for repair or maintenance.
4. Power lost in cable depends on ……, ……, …….
5. The …….. can be reduced by the application of a shunt capacitor.
6. In a tape changing transformer, the tapping are provided on ……. voltage winding.
7. Modern power system consists of four main subsystems: …….., …….., ……., and ……..

8. A …… is an electronic device used to issue tripping commands to the corresponding circuit breakers in order to interrupt the current produced by the fault.
9. ……. the size of conductor is one way to lower cable power losses.
10. The recloser opens and recloses ……. or ……. times.
11. The voltage of the single-phase supply to residential consumers is _____.
12. There are four main functions of the distribution substation ……, ……., ……, ……..
13. In ……. phase, electric power is moved to individual or group of loads.
14. Transformer is……
15. Distribution system element such as …….., …….., ……, …….
16. …… is a communication system that allows real-time monitoring of the distribution system.
17. …… is a device that can only carry and interrupt normal load current and thus disconnect portions of the network.
18. A feeder in transmission system, feed power to ……..
19. The ……. distribution system provides electric energy to the customers.
20. The Common configurations of primary arrangements with a single feeder are ……. mainline, or the …… mainline.
21. The optimum location of the capacitor bank is around …….. of the length of the feeder.
22.The capacitor bank …….. and reduces voltage drop at heavy loads.
23. The most frequently used voltages and wiring in the primary distribution are ……., or …… in Egypt.
24. The distribution systems has two parts, …….. and ……..
25. The most frequently used voltages and wiring in the Primary distribution are ……. kV in Egypt.
26. A capacitor bank …….. power factor.
27. The problem with …… connection is the Circulating current that occurs when the two supply station voltages are different.
28. Shunt capacitors are used for ……. and ……..
29. At light loads, the capacitor is switched off to avoid ……..
30. The active losses can be simplified to ……..., The reactive power loss can be simplified as …….
31. The power factor of an electrical power system is in the closed internal of ……..
32. The apparent power is …… the real power When the power factor = 1.
33. …… is the capacity of the circuit for performing work in a particular time.
34. The values of the power losses depend on ……..
35. When the power factor = ……., the voltage and the current are in phase.
36. The capacitor is switched …. during the night when the load is …….. and switched …. when the load is ……
37. In industrial facilities ……. Load include incandescent lights, and ……. load include AC induction motors.
38. A power factor of ……. means that the voltage and current waveforms are not in phase.
39. Shunt ……. are used for power factor correction.
40. The apparent power is …… than the real power. When the power factor less than 1.
41. The most frequent use of capacitors is on the …….. feeders.
42. The ……. is the sum of continuous ratings of all the equipment's connected in supply system.
43. ……. is the load that is drawn from the supply at a specified Interval of time; it is expressed in KW's, KVA or Amperes.
44. ……. is a graphic record showing the demand of the power for every instant during the hour, the day, the month or the year.

45. The load factor is the ratio of the average load to the …….. load for a certain period of time.
46. ……. is the time that an equipment is in use / The total time that it could be in use.
47. The …… is the average power that consumed by the load during a certain period of time.
48. The electricity is most often generated at a power station by ……..
49. The voltage drop (ΔV) is defined as ……
50. The feeder power losses are of two main different kinds as …….. and ……...

Q3. Put (✓) on true sentence and (X) on false sentence:-

1. A relay is an electronic, low-powered device used to activate a high-powered device.
2. Shorten the length of the cable is one way to lower cable power losses.
3. The electricity is most often generated at a power station by batteries.
4. The power lost in a cable depends on the cable length cable size and current through the cable.
5. A Switch is a switching device used to isolate a system element for repair or maintenance.
6. Distribution is the last section of the electrical power system operated by the utility grid.
7. One way to lower power losses, using shorten length cables.
8. Relay is responsible for transporting electrical energy between two distant points.
9. In current transformers, the primary winding is connected in series with the circuit carrying the line current.
10. Recloser devices similar in function to circuit breakers, except they also have the ability to reclose after opening.
11. The ability of iron or steel to carry magnetic flux is much less than air.
12. Transformer transfers power from one winding to another with changing the frequency.
13. The voltage supplied to each customer on a distribution feeder should be within ± 10% limit.
14. The capacitor is switched on when the load is light and switched off when the load is heavy.
15. A capacitor bank is a local source of active power.
16. The load factor is the ratio of the connected load to the maximum load for a certain period of time.
17. The apparent power is lower than the real power.
18. Primary distribution, in low load density areas, is a radial system. This is very expensive but yields low reliability.
19. The power factor cos Ø= P/S
20. The area under daily load curve is greater than area under the corresponding daily load duration curve.
21. The load factor of a power plant more than unity.
22. The demand factor is the ratio of the average load to the connected load.
23. The general radial circuit design can present such as the single mainline, or the branched main line.
24. The real power is lower than the apparent power. When the power factor < 1.
25. Active power loss can be simplified as I^2X.
26. The apparent power is higher than the real power.
27. Operating value of diversity factor Greater than unity.
28. Resistive loads include incandescent lights, and Inductive loads include AC induction motors.
29. The power factor of industrial loads is generally Lagging.
30. Demand factor is Maximum demand to connected load.
31. The average load is the average impedance that consumed by the load during a certain period of time.
32. The maximum value of power factor equal 1.
33. Primary distribution, in low load density areas, is a radial system.

34. A three-phase switched capacitor bank is rated two-thirds of the total average reactive load and installed two-thirds of the distance out on the feeder from the source.
35. At the light load condition, the voltage supplied to each customer will be within the allowable limit.
36. In an electric power system, a load with a low power factor draws less current than a load with a high-power factor for the same amount of useful power transferred.
37. Auto transformer is an electrical transformer with only one winding.

Q 4: Define: Fuse - Circuit Breaker - Diversity factor - Potential transformer - Sectionalizer - Capacitor Bank - Voltage regulator - Demand Factor – Power losses - Primary distribution system - Advantages of improving the power factor - current transformer - load factor - power factor - secondary distribution system - Resistive loads, Inductive loads and capacitive loads.

Q 5: Compare:

1. Circuit Breaker, Fuse and switch.
2. Active power loss, Reactive power loss.
3. Power Loss, Power Factor.
4. Demand Factor and Diversity factor.
5. Lagging power factor, leading power factor.

Q6. Match between column (A) and (B):

1. Lines are responsible for	A. Devices that can carry and interrupt normal load current and thus disconnect portions of the network.
2. The sub transmission system connects	B. an underground cable system is used.
3. Switches	C. The high-voltage substations to the distribution substations
4. In urban areas	D. consists of a strip of wire that melts and clears an electric circuit when an overcurrent passes through it.
5. A fuse	E. Transporting electrical energy between two distant points

1. Recloser	A. an electronic, low-powered device used to activate a high powered device.
2. A relay	B. is a local source of reactive power. By correcting power factor it can perform voltage regulation and reduce system losses.
3. A capacitor bank	C. Devices that can carry and interrupt normal load current. In addition, they interrupt short-circuit (fault) current.
4. Circuit breakers	D. is a circuit-opening device used in conjunction with source-side protective devices.
5. Sectionalizer	E. Devices have the ability to reclose after opening, open again, and reclose again, repeating this cycle a predetermined number of times until they lockout.

Q7: Solve:

A wound coil that has and inductance of 120 mh and a resistance of 25 Ω is connected in series with capacitance 100 µf , are connected to a 200 volt 50 hz supply calculate:
The impedance of the circuit - The current - The power factor - The apparent power consumed - Draw the resulting power triangle for the circuit.